Fiscal Year:	FY 2013	**Task Last Updated:**	FY 01/02/2013
PI Name:	Rickus, Jenna Ph.D.		
Project Title:	Sporesat Mission: Investigating biophysical mechanisms of plant gravisensing using a lab-on-a-chip approach		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Cell Culture (2) Bioregenerative Life Support		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 SALMON--Astrobiology & FSB NNH08ZDA009O-SCMAFSB
Start Date:	03/01/2011	**End Date:**	09/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	Small Satellites		
Flight Assignment:	NOTE: Extended to 9/30/2014 per NSSC information and A. Chu/NASA ARC (Ed., 1/6/2014)		
Key Personnel Changes/Previous PI:	Previous PI: Marshall Porterfield. PI change to Jenna Rickus in November 2012. Amani Salim replaced him initially until November 2012 and she remains as Co-PI.		
COI Name (Institution):	Salim, Amani (Purdue University/ Co-Principal Investigator) Roux, Stanley (University of Texas, Austin)		
Grant/Contract No.:	NNX11AF48A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A detailed understanding of the molecular and biophysical mechanisms for plant gravity sensing still eludes the scientific community. To gain deeper knowledge of the mechanism of cell gravity sensing, we will study activation of cell gravisensing and electrophysiological signaling in a single-cell model system (Ceratopteris richardii spores) using a "lab-on-a-chip" microsensor technology platform. We here propose a Small Complete Mission (SCM) based on a Mission of Opportunity-2 nanosatellite we call SporeSat, which will adapt hardware and technologies from previous and current missions including: GeneSat, PharmaSat, O/OREOS and GraviSat. Our approach allows for real-time measurement of calcium signaling and maintenance of variable gravity treatments during the flight experiment. Our science objectives further current work that has documented molecular components of biomechanical transduction. Our technical objectives focus on refining our current lab-on-a-chip sensor system to build a new CEL-C Advanced bioCD designed for the micro-satellite platform. This project contributes to NASA's initiatives in the areas of both Astrobiology and Fundamental Space Biology and will advance the science and technology capabilities for future research and development of astronaut health programs as well as those for advanced life support for exploration in the space environment, the moon, and on other planets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Overview: SporeSat goal is to investigate the role of calcium ion in fern spore Ceraptopteris richardii gravity sensing mechanism. In previous ground and C9 flight experiments, C. richardii extracellular calcium current was found to be polarized by gravity and the polarization has effect on its subsequent nuclear migration and rhizoid development. However, the onset of the polarity development is still unknown. To elucidate the role of calcium ion polarization on C. richardii fern spore development, we have developed a centrifuge microsensor platform that could provide various artificial gravity levels to the fern spore in microgravity, called SporeSat. SporeSat houses a sensor unit called the bioCD – a 2 inch diameter disc that has multiple sensing microelectrodes. At various rotation rates, spores at four radial distances from the center of the centrifuge platform experienced different levels of artificial gravity. Due to the variable gravity conditions, the onset of polarity development can be determined. SporeSat is a close collaborative effort between the Purdue University Physiological Sensing Facility (PSF) and the Ames Research Center (ARC) Small Spacecraft Payloads and Technologies (SSPT). We have clearly defined the roles and responsibilities of Purdue and ARC and the collaboration has made significant progress on the project. The bioCD lab-on-a-chip system is the product of several areas of research and development, which include 1) sensor research and device fabrication, 2) sensor packaging that is spaceflight compatible, 3) amp board design and development, 4) application testing on fern spore C. richardii, 5) spacecraft hardware development, and 6) risk analysis (i.e., shock and vibration, biocompatibility, spore dislodge test). Purdue is responsible for items 1, 2, and 4; ARC is responsible for item 5; and both Purdue and ARC are responsible for items 3 and 6. This report describes the status of the items for which Purdue has some or all responsibility. Sensor research and device fabrication: A lab-on-a-chip electrochemical sensor named the bioCD sensing disc has been designed: a 2-inch fused-silica disc that consists of 32 laser-etched wells, each 160 µm in diameter. In each well, a pair of all-solid-state ion-selective electrodes that are selective for calcium ion are patterned at each flat side of the etched well. Each well is contained in a microfluidic chamber that holds nutrients for the fern spore. A complete bioCD sensing disc has been developed; the calcium ion recognition layer is currently being tested and optimized. A pair of contact pads connected to each electrode provides electrical connection via pogo pins to an amplifier board. Each sensing electrodes has been calibrated by conducting potentiometric measurements in known calcium concentration. The sensor performance is currently being optimized for detection limit and linear range. Sensor packaging that is spaceflight compatible: A packaging unit to contained the bioCD sensing disc is comprised of two parts: a bioCD enclosed assembly that consists of a microfludic chamber housing the bioCD sensor unit, and a bioCD rotating assembly that holds the bioCD enclosed assembly and is sealed from outside environment. The materials of the bioCD assemblies were chosen to be flight compatible with low degassing properties. Gaskets are used to provide additional shock protection and electrical insulation to the whole bioCD assembly. Lighting condition for C. richardii fern spore germination: The germination of the fern spore C. richardii is initiated by red light at a set intensity. Uniform lighting condition is important to provide simultaneous germination of the 32 fern spores in each bioCD. We tested germination with optimized lighting condition; microscope inspection showed that fern spores germinated uniformly. Amp board design and development: An amplifier PCB was designed, fabricated, and tested for electrical functionality. A total of 32 amplifiers amplify the differential voltage signal from the pair of sensing electrodes in each bioCD well. Signal from each electrode pair goes through an A/D converter, where analog signals are digitized. The digitized signals from the 32 sensing electrodes pairs go through a multiplexer that compresses all the signal into a single channel. Finally, amplifier stability was verified and optimized. We are in the process of integrating the amplifier PCB with the bioCD packaging unit and currently performing electrical measurements to test electrode stability. We are also resolving the issue of bias current in the circuit by optimizing the transducer layer. Preliminary analysis results are positive in terms of project viability. This testing will allow us to optimize the amplifier PCB to amplify the differential voltage signals from the sensing electrode pairs, and meet the specifications of the science requirements. Application testing on C. richardii fern spores: We will initiate testing on C. richardii fern spores after the optimization of the sensing electrodes and amp board as stated above. This is planned to occur before the critical design review (CDR), which is scheduled to be held in the second week of February, 2013. Risk analysis: Shock and vibration environment expected during launch on each spore location in bioCD sensing disc was simulated. The calculated maximum stress is less than the tensile strength of bioCD sensing disc material. Using frequency response and modal analysis, the shock and vibration test condition performed at ARC was simulated. The modeling result will be used to optimize brace supporting design after being integrated with spore dislodge test result. This optimization will relieve shock and vibration and reduce risk of bioCD mechanical failure. Initial shock and vibration tests were conducted on fern spores in bioCD wells for 3 bioCD rotating assemblies. The bioCD assemblies and sensing disc survived the shock and vibration test. We have perfected the spore-loading procedure to ensure that 99% of spores are not dislodged from the wells after experiencing shock and vibration. The

	second dislodge test is scheduled for sometime in early January, 2013. The biocompatibility protocol is based on previous missions. Currently, we are preparing bioCD sensing discs and assemblies for the test, which is scheduled for sometime in mid January, 2013. Summary of research and development: A prototype of a bioCD sensing disc and its packaging have been completed and is currently being fabricated for a second dislodge test/shock and vibration test, biocompatibility tests, and ground flight-simulation tests. These units are fabricated from flight-compatible materials and have survived shock and vibration tests conducted at ARC. The bioCD sensing disc recognition layers have been optimized and tested, resulting in a stable signal when amplified. Lighting conditions for spore germination have been optimized by the use of an OLED to replace the non-uniform lighting from surface-mounted LEDs. The bioCD packaging unit is currently being tested for leakage, and measurements are currently being conducted with fern spores at 1g.
Bibliography Type:	Description: (Last Updated: 04/26/2017)
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